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Ans: D

Using potential divider concept,

$$\begin{aligned} \text{e.m.f. of cell} &= \text{p.d. across the balance length of wire} = E \left(\frac{\frac{1}{3}R}{r + 3R + r} \right) \\ &= \frac{ER}{3(4R + r)} \end{aligned}$$

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Ans: D